Filipe Nascimento

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Summary

I have a CS Ph.D. and am passionate about problem-solving, computer graphics, geometry processing, and physically based simulation. My experience includes the design/implementation of fluid solvers, real-time rendering techniques, GPU programming, geometric and spatial data structures, and physically based animation research.

Education

PhD in Computer Science 2017 - 2024
M.S. in Computer Science 2013 - 2016
B.S. in Computer Science 2008 - 2012

Institute of Mathematics and Computer Science (ICMC) -
University of São Paulo (USP), São Carlos, São Paulo, Brazil

Skills & Interests

OpenGL/CUDA/Houdini's HDK/OpenVDB/OpenUSD/OpenFOAM
Vulkan/OpenSubdiv/M. Learning

Programming

C/C++(proficient)/Python *and some experience with*: MATLAB/Rust/R

Languages

Portuguese, English, French (Débutant), Japanese (Beginner)

Professional Experience

RnD Software Engineer at Blizzard Entertainment Jan 2023 - Nov 2024
Worked on C++ 3D tools for cinematics at Blizzard Animation:
- Optimization and extension of the proprietary hair system, including the re-design of data structures, research of interpolation methods, and implementation of visual debugging tools;
- Houdini and Katana plugin C++ development;
- Integration of OpenUSD into existing code.

Software Engineering Intern at Google Inc. (YouTube) Jan - Mar 2016
Worked on 360^o video support for YouTube app on Sony's PlayStation 4 VR.

Research Experience

Digital Animation of Powder Snow Avalanches 2017 - 2022
Applied CFD to solve geophysical equations to simulate avalanches and produce animation data. Developed in C++, including all geometric processing.
Graduate Research supported by FAPESP

Multimaterial Fluid Simulation for Comp. Graphics Sep 2014 - Feb 2015
Visiting Scholar at University of Waterloo (UW), Canada. Worked with dynamic polygonal surfaces, continuous collision detection, and tet meshes.
Supervisor: Christopher Batty

Reliable polygonal approximation of implicit curves 2011 - 2012
Undergraduate Research supported by FAPESP. Supervisor: Afonso Paiva

Scientific Publications

Digital Animation of Powder-Snow Avalanches 2020
Paper at ACM Siggraph 2025 and ACM Transactions on Graphics

RBF Liquids: An Adaptive PIC Solver Using RBF-FD 2020
Paper at ACM Siggraph Asia 2020 and ACM Transactions on Graphics

Approximating implicit curves on plane and surface triangulations with affine arithmetic (AA) 2014
Paper at Computers & Graphics Journal (CAG), Volume 40

Approximating implicit curves on triangulations with AA 2012
Paper at XXV SIBGRAPI Conference on Graphics, Patterns and Images

Activities

Programming Contests (ACM-ICPC)
- World Finals (as coach) 2014
- Latin America Regional Contest (1st Place) (as coach) 2013
- Latin America Regional Contest 2009
- Brazilian Regional Contest 2009 - 2012

Teaching Assistant
- Advanced Algorithms Laboratory Course 2013